Individual Contribution Statement

Section 1. Work Division Table

Part	Haixiao Yang	Xiaoli Wu	Yian Zhou	Notes
(1) DQR(25/100)	40%	10%	50%	Haixiao:data understanding code and markdowns Xiaoli: group discussion Yian: PDF report
(2) DQP(15/100)	10%	40%	50%	Haixiao: group discussion Xiaoli: data cleaning code and markdowns Yian: PDF report
(3) Feature pairs (15/100)	80%	10%	10%	Haixiao: feature pairs code and markdowns Xiaoli: group discussion Yian: group discussion
(4) New features (15/100)	10%	80%	10%	Haixiao: group discussion Xiaoli: new features code and markdowns Yian: group discussion
(5) Models (30/100)	35%	35%	30%	Haixiao: train-validation date split, RidgeCV and test data processing Xiaoli: Random forest, Pipeline Yian: Linear regression, Lasso
Overall	**35.5%**	**32.5%**	**32%**	

Section 2. Detailed Task Log

Name: Haixiao Yang		
Part	**Task**	**Time (Hours)**
(1) DQR	Implement data understanding code	3
	Provide explanations and justification for data understanding	1
(2) DQP	Discuss with group members about the strategy for DQP	0.5
(3) Feature pairs	Implement feature pairing code and explanation	3
	Provide explanations and justification for feature pairing	1
(4) New features	Discuss with group members about DQP	0.5
(5) Models	Implement train-validation data split	1
	Implement RidgeCV model	2
	Analyze model performance (RidgeCV)	1
	Processing test data set with the feature engineering steps	2
	Implement the evaluation with clamped test date set	1

Name: Xiaoli Wu		
Part	**Task**	**Time (Hours)**
(1) DQR	Discuss with group members about potential data quality issues and proposed solutions	1
(2) DQP	Implement data cleaning code	3
	Provide explanations and justification for data cleaning steps.	1
(3) Feature pairs	Discuss with group members about feature pairing result	0.5
(4) New features	Implement new features code and explanation	3
	Provide explanations and justification for new features.	1
(5) Models	Design and implement the Random Forest model	3
	Analyze model performance (Random Forest model)	0.5
	Design and implement a pipeline	2
	Analyze pipeline model performance	0.5
	Interpret and explain the differences between training and test performance results	1

Name: Yian Zhou		
Part	**Task**	**Time (Hours)**
(1) DQR	Report writing	3
	Update report with comments	0.5
(2) DQP	Report writing	4
	Update report with comments	0.5
(3) Feature pairs	Discuss with group members about feature pairing result	0.5
(4) New features	Discuss with group members about new features	0.5
(5) Models	Design and implement the Linear regression model	2
	Analyze model performance (Linear regression)	1
	Design and implement the Lasso model	2
	Analyze model performance (Lasso)	1

Section 3. Signed Declaration

I certify this contribution statement is accurate. I understand that misrepresentation is academic misconduct.

Signature: Haixiao Yang, Xiaoli Wu, Yian Zhou
Date: March 29th, 2026